

0:00	Title	“Good morning. Martin Maxa, CTU Prague, with Dr. Jakub Svatoš.”
0:30	Why	shootings up, ownership up camera needs line of sight, sound does not
1:15	Signature	shock wave first ≈ 2.5 ms, only if $v > c$ [CLICK] muzzle blast ≈ 48 ms between = reflections (comes back later)
2:30	Naive	spectrum = environment time domain = dispersion $\Rightarrow$ need both
3:00	System	stand-alone units $\rightarrow$ server classify + 2D coverage & two bearings
4:00	Localize	2 MEMS, $d = 0.186$ m, $f_s = 44.1$ kHz $\Theta = \arcsin(c \Delta n / (d f_s))$
5:00	Features	[CLICK] two units cross MAE 0.77°, 70 events frame $\rightarrow$ FFT $\rightarrow$ power $\rightarrow$ log $\rightarrow$ DCT filter bank is what differs MFCC / LFCC / IMFCC / GTCC (gammatone = the ear) 3 frame lengths $\times$ 2 classifiers
6:15	TABLE	rows = frame length, columns = ACC / RLC / MCC look only at the red rows — best in every block best: GTCC + NN, 50 ms $\rightarrow$ 99.15 / 96.77 / 0.98 context: 117 of 118 correct, recall 30 of 31
7:45	Numbers	mel-family all the same GTCC always wins NN > SVM, gap grows longer frames better <i>but</i> start hearing the room
8:45	Conclusion	network cepstral works GTCC > mel, NN > SVM real firearms
9:30		“Thank you — I am happy to take questions.”

If asked, the numbers: dataset 371 train / 118 test, five classes | 9 mm: 70 / 31 | 26 features, 75/25 split | NN: Levenberg–Marquardt, 370 iterations | bearing: cross-corr 0.84°, +parabolic 0.77°, PBDE 3.24° | backup slides 11–14

Pace: if you reach the results table after 6:30, skip the second bullet on “What the numbers say”.